

Problem Set 4
Due: Mar. 22, 2021

1. Section 2.2 Problem 4.

An internet service provider uses 50 modems to serve the needs of 1000 customers. It is estimated that at a given time, each customer will need a connection with probability 0.01, independent of the other customers.

- (a). What is the PMF of the number of modems in use at the given time?
- (b). Repeat part (a) by approximating the PMF of the number of customers that need a connection with a Poisson PMF.
- (c). What is the probability that there are more customers needing a connection than there are modems? Provide an exact, as well as an approximate formula based on the Poisson approximation of part (b).

2. Section 2.3 Problem 16.

Let X be a random variable with PMF

$$p_X(x) = \begin{cases} x^2/a & \text{if } x = -3, -2, -1, 0, 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a). Find a and $\mathbf{E}[X]$.
 - (b). What is the PMF of the random variable $Z = (X - \mathbf{E}[X])^2$?
 - (c). Using the result from part (b), find the variance of X .
 - (d). Find the variance of X using the formula $\text{var}(X) = \sum_x (x - \mathbf{E}[X])^2 p_X(x)$.
3. Suppose that two teams play a series of games that ends when one of them has won 3 games. Suppose that each game played is, independently, won by team A with probability p . Find the expected number of games that are played. Also, show that this number is maximized when $p = 1/2$.

4. Let X be a RV such that

$$\begin{cases} p_X(1) = p \\ p_X(-1) = 1 - p \end{cases}$$

Find $c \neq 1$, such that $\mathbf{E}[c^X] = 1$.